

Measuring phosphorus fluxes through the root zone of a layered sandy soil: comparisons between lysimeter and suction cell solution

J. MAGID*, N. CHRISTENSEN & the late H. NIELSEN

Division of Terrestrial Ecology, National Environmental Research Institute, Silkeborg, Denmark

SUMMARY

Chemical and physical processes govern the extent to which phosphorus dissolved from the plough layer is retained by or leached from the subsoil. The adsorption capacity of deeper layers, the velocity of solution flux and the amount of soil volume in contact with the leaching soil solution are important determining factors.

Soil solution from 90-cm depth of a layered sandy soil, sampled over a 3-year period with teflon suction cells, was systematically lower in P content than soil solution sampled with free drainage lysimeters from the same depth. On average 4.6 times as much inorganic P and twice as much organic P was found in lysimeter solutions.

Determination of apparent equilibrium solution concentrations of phosphate was performed by means of a series of batch experiments with soil and soil solution of varying initial P content. The results indicated that the suction-cell solution was close to equilibrium with soil from the 90-cm layer.

Rhodamine dye was applied, and flow paths of soil solution observed. Preferential flow was initiated from layers of finer texture overlying layers of coarser texture.

On the basis of these observations it is concluded that suction cells give the best representation of stagnant soil solution, whereas lysimeters give a better representation of mobile soil solution.

INTRODUCTION

Phosphorus in its inorganic and known organic forms has a high affinity for the soil matrix, and it is generally assumed that the extent of phosphorus leaching from soil is small. In the literature there are few reports contrary to this. As a control of excessive algal growth, inorganic P in surface waters with moderate to high biological activity should not exceed $1\text{ }\mu\text{M}$, and total P concentration should stay below $2\text{ }\mu\text{M}$ (Taylor & Kilmer, 1980). In drainage water, inorganic P concentrations of around $6\text{ }\mu\text{M}$ have been reported (Bolton *et al.*, 1970; Sharpley & Menzel, 1987). Schofield & Birch (1986) studied the sources of eutrophication of the Peel-Harvey estuary. Water from agricultural drains going directly into the estuary had inorganic P concentrations in the range of $39\text{--}81\text{ }\mu\text{M}$, but shortly after the spring fertilization in the area, the inorganic P concentration reached $267\text{ }\mu\text{M}$. Ulén (1986) demonstrated, for a number of Swedish soils, a high correlation ($r=0.95$) between the amount of P that could be leached with distilled water from plough-layer samples in the laboratory, and the median total P concentrations in drainage water from the soils during a 3-year period. Thus a close relationship between drainage water P and plough-layer content of soluble P was suggested.

Rain-water residing in the plough layer for a short time will be enriched with phosphorus to an

*Present address: The Royal Veterinary and Agricultural University, Thoraldsensvej 40, 1871 Frederiksberg C, Copenhagen, Denmark.

extent that reflects the P status of the soil. In order that the phosphorus is sufficiently re-adsorbed by the soil before it reaches aquatic systems, the water must pass reactive surfaces for long enough to allow the concentration to fall below the critical level of $1\ \mu\text{M}$ inorganic P as estimated by Taylor & Kilmer (1980). The reactions of phosphate with soil are complex, and it is questionable when equilibrium between phosphate in soil and in the soil solution is attained. Beckett & White (1964) described a method for evaluating the quantity/intensity relation for phosphate sorbed to soil vs. phosphate in solution, and observed that the equilibrium between soil and solution was time dependent in an Upper Greensand field soil.

In recent years there have been several reports concerning preferential flow, or by-pass flow, of soil solution in structured soils (Schaffer *et al.*, 1979; Brown *et al.*, 1985; Richter & Jury, 1986; Priebe & Blackmer, 1989). Schaffer *et al.* (1979) demonstrated that nitrate concentration in water percolating at a depth of 1.2 m was closely related to the concentration of nitrate in irrigation water, whereas water sampled with suction cups showed rather small variations in nitrate content. Evidence of preferential flow in sandy (less structured) soil is more scarce. In their work on infiltration theory, Tamai *et al.* (1987) demonstrated that unsaturated flow of both water and silicon oil through a homogeneous porous medium would result in fingering, because pressure differences created laterally, due to the air/water interfaces, made liquid converge into fingers. Kung (1990) followed the flow paths of Rhodamine Red WT, a slightly reactive strong dye, to a depth of 7 m in a sandy soil. At a depth of 1.5 m, less than 50% of the soil was coloured, 10% at 3.0 m depth and 1% at 6 m depth. Thus there is adequate documentation that preferential flow may be important in sandy soil, and that an unverified assumption of piston flow of water through sandy soil may be inappropriate.

We initiated an investigation of ion fluxes in a field experiment in order to obtain a balance of nutrient input and output for several vegetational types (Magid & Christensen, unpublished data). We employed several methods for collecting soil solution, and observed systematic differences in solution composition, particularly with regard to phosphorus content. In this paper we focus on these methodological differences, in an attempt to explain the underlying mechanism and to evaluate the applicability of the methods to soil-solution studies.

MATERIALS AND METHODS

Sampling methods

Soil solution was sampled at a depth of 90 cm in two ways, (i) with teflon suction cells (Prenart, Denmark) and (ii) with free drainage lysimeters. The teflon suction cells were used because they only sorb small amounts of P (Beier *et al.*, 1989), in contrast to ceramic suction cells. Ten suction cells were installed following the guidelines of Rhoades & Oster (1986), and each cell was connected to a 1-dm³ borosilicate flask that was evacuated to a pressure of about 0.03 MPa for sampling soil solution. In the same plot we installed a further six lysimeters made of perspex tube (20 cm in diameter, 50 cm deep) attached to a perforated plate and a funnel (Fig. 1). An undisturbed soil volume for filling the lysimeters was sampled with a stainless steel core of similar proportions, from a depth of 40 cm to 90 cm. From the bottom plate of the lysimeter, wicks (15 cm) were drawn through a funnel to enhance drainage into the sampling jar (6 dm³). The lysimeters were then installed with the bottom plate at 90 cm depth and the topsoil was carefully re-established. The lysimeters and suction cells were sampled every 2 weeks during the wet season.

Soil was collected at the site in St Lyngby, North Zealand, Denmark, using an auger 4 cm in diameter. The St Lyngby soil has developed from windblown sand on top of alternating layers of alluvial sand and silt. Sixty cores were collected to a depth of 90 cm. From these cores the plough layer, the 30–60 cm and the 60–90 cm layer were separated and then bulked to give one composite sample representing each layer.

Chemical analysis

The soil solutions were analysed for Cl, NO₃ and SO₄ ions by anion chromatography, Ca, Mg, Fe and Al by atomic absorption spectroscopy (AAS) and K and Na by flame emission spectroscopy (FES). Inorganic phosphate was determined as molybdate-reactive P by the method of Murphy &

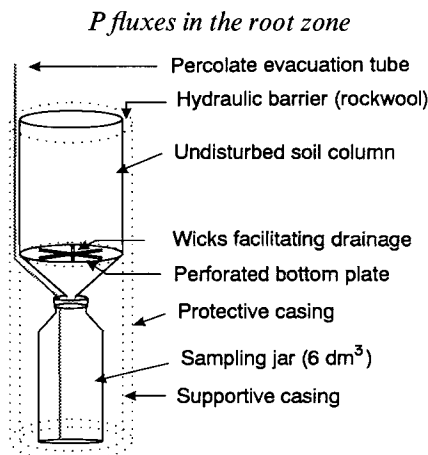


Fig. 1. Schematic diagram of the lysimeter construction.

Riley (1962), and organic phosphorus was determined as the difference between molybdate-reactive P before and after autoclaving the samples in a mixture of NaOH and $K_2S_2O_8$ (140°C for 20 min) by the method of Ebina *et al.* (1983).

Analysis of soils included: NH_4 -acetate exchangeable Ca, Na, K, Mg and exchangeable acidity according to Thomas (1982), texture, pH, and oxalate-extractable Fe and Al by extraction in darkness for 4 h at pH 3 (McKeague & Day, 1966).

Apparent equilibrium phosphate concentrations were determined using a modification of the procedure suggested by Beckett & White (1964). Based on the work of Barrow (1987), it was decided that in order to obtain a realistic estimate of the stagnant water concentration of phosphate in the plough layer and the 60–90 cm layer, it was necessary to study the changes over time and at different solution:soil ratios. An electrolyte solution containing Na, Ca, Mg, SO_4 , Cl and NO_3 , at concentrations similar to those found in the soil (unpublished data), was prepared and the pH adjusted to 7.0. K_2HPO_4 was added to give concentrations of 0, 2, 6, 15, 50, 100 and 2000 μM P. At solution:soil ratios of 2 and 10, duplicate soil samples were shaken in the above solutions. The reaction of phosphate with soil was monitored by sampling the mixture over time for 48 h. Samples were centrifuged (6000 g), and molybdate-reactive P was determined in the supernatant. A characteristic plot of P concentration vs. time is given in Fig. 2. Depending on the initial level of phosphate in the batch solution and the time of reaction, a net adsorption or desorption occurred. By plotting the initial concentration vs. the net desorption or net adsorption, the apparent equilibrium concentration may be determined as the concentration at which neither desorption nor adsorption occurred (Fig. 2).

Application of dye

In order to gain an impression of the vertical water movement, Rhodamine dye, as used by Kung (1990), was applied to 1-m² patches of soil with a sprayer. The amount applied was approximately 2 mm of a 0.1% dye solution. The movement of dye was followed by removing soil after 2 and 4 months.

Data analysis

Graphical and statistical analysis of data was conducted using the SAS software, particularly the GLM and GPLOT procedures (SAS Institute Inc., 1989, 1991). Flux-averaged concentration, c_f , was calculated according to

$$c_f = \frac{\sum_{n=1}^i (c_n q_n)}{\sum_{n=1}^i q_n}$$

where c_n is the concentration in lysimeter n and q_n is the volume of water contained in lysimeter n .

RESULTS AND DISCUSSION

The results of the chemical and physical analysis of the soils are given in Table 1. The texture changed from coarse to fine with depth. Thus the plough layer had the largest amount of coarse sand, and the deepest layer contained the largest amount of clay. This does not, however, reveal the soil heterogeneity to its full extent as several thin, distinct (1–3 cm) layers with a pseudogley character were found from 30–80 cm depth during the wet season. Based on the content of amorphous iron and aluminium oxides, the adsorption capacity for P was estimated to be approximately 3.8, 2.4 and 3.3 $\mu\text{mol P g}^{-1}$ soil for the plough layer, the 30–60 cm layer and the 60–90 cm layer, respectively (Borggaard *et al.*, 1990). The contents of resin-extractable P and exchangeable Ca show the effect of liming and fertilization on the plough layer. This is also reflected in the $\text{pH}(\text{H}_2\text{O})$ of the layers that are in the neutral–alkaline range.

Apparent equilibrium points were determined graphically as described in Fig. 2, and the results are shown in Table 2. Given the uncertainty of the determination, there is no reason to believe that the solution:soil ratio significantly affected the apparent equilibrium concentration of the plough-layer soil. This is in agreement with observations of Beckett & White (1964). After 2 h, a concentration of about 40 μM P was reached. It is therefore reasonable to extrapolate to the much lower solution:soil ratio (0.1–0.2) found under field conditions.

For the 60–90 cm layer the apparent equilibrium concentrations were lower when the solution:soil ratio was lower. As the amount of P in solution is greatest in the high solution:soil ratio experiment, this implies that the adsorption of P is rate-limiting for the attainment of equilibrium under the experimental conditions used. Although it can be assumed that the apparent equilibrium concentrations should be even lower under field moisture conditions, extrapolation cannot be performed. However, it is clear that within a time span of 48 h the apparent equilibrium concentration is decreasing in the 60–90 cm layer. Accordingly, the equilibrium concentration of stagnant water in this layer should be less than 2 μM ; this figure is regarded as a maximum equilibrium concentration for stagnant solution hereafter.

Soil solution phosphate concentrations

Figure 3 shows the concentrations of inorganic P in the sampled soil solution isolated by suction cells and lysimeters during three drainage periods in the years 1987–1990. The flux-averaged concentrations and the arithmetic means and average standard errors of means have been shown for both kinds of solution. It can be seen that the concentrations during the first 3 months of the initial drainage period were quite similar. Later on there was no similarity, as the suction-cell concentration of P remained below or rather close to the above estimated maximum concentration of 2 μM P for stagnant water in the 60–90 cm layer. The concentrations in lysimeter solution later on were higher than the initial concentrations, and far above the estimated maximum stagnant water concentration. There was a tendency for the flux-averaged concentration in lysimeter solution to be higher than the arithmetic mean values. This is because there was a weak but significant positive correlation between the amount of water in the lysimeter flask and the concentration of P ($r^2 = 0.1224$, $p > 0.01$). In all drainage periods the flux-averaged concentration of inorganic P in lysimeters increased towards the end of the drainage period.

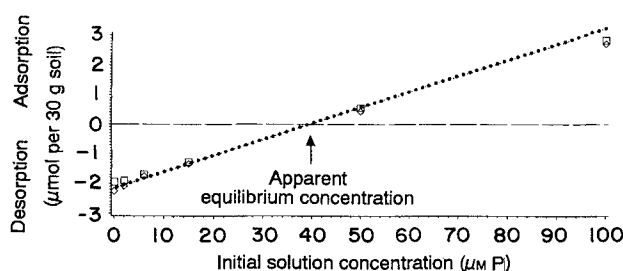
Chloride is often regarded as a non-adsorbed ion in soils where the amounts of positively charged sesquioxides are small. The amounts of oxalate-extractable iron and aluminium, the neutral to slightly alkaline pH values in the soil, and the actual chloride concentrations, suggest negligible chloride adsorption. In Fig. 4 the ratio of the chloride flux concentration in lysimeter solution to that in the suction-cell solution is plotted against time. Each drainage season showed a pattern, whereby the content in the lysimeter solution was highest at the beginning of the season, and lower later on. During the growing season the chloride concentration in the soil solution increased because of water evaporation. Thus a chloride peak is expected at the beginning of the drainage period. The patterns in Fig. 4 indicate that this water reached the lysimeters earlier than the suction cells, and that the suction cells sampled soil solution with a higher chloride content for a prolonged period.

Organic phosphorus concentration (Fig. 5) decreased from the beginning of the drainage season in the first two periods, and increased towards the end of the drainage season in all periods. This

Table 1. Selected properties of the soil studied

	Particle-size distribution (g kg ⁻¹)					pH in water	Fe ^a	Al ^a	Ca ^b	Resin P ^c
	< 2 µm	2–20 µm	20–63 µm	63–200 µm	200–2000 µm					
Plough layer	41	38	66	620	225	6.9	11.7	11.1	31.6	1.02
30–60 cm	42	24	93	331	508	7.4	9.1	6.3	19.2	0.48
60–90 cm	103	97	157	569	72	8.2	14.3	7.4	39.9	0.51

^aOxalate-extractable Fe and Al (µmol g⁻¹ soil); ^bNH₄-acetate exchangeable Ca (µmol g⁻¹ soil); ^c(µmol g⁻¹ soil).

**Fig. 2.** Determination of apparent equilibrium concentration for a given reaction time (6 h) at a given solution: soil ratio (2).**Table 2.** Apparent equilibrium concentration of inorganic P as a function of time and soil:solution ratio

		Time (h)						
		Solution:soil ratio	0.5	1.5	2.5	4	6	23
Plough layer	2:1	27	32	40	38	37	48	36.1
Plough layer	10:1	27	42	39	39	37	40	38.1
60–90 cm	2:1	2.6	2.3	2.6	2.3	1.9	1.3	1.3
60–90 cm	10:1	4.2	4.2	5.2	4.2	2.9	2.9	1.9

Mean values (µM P) with coefficient of variation in the range of 6–30%.

pattern probably reflects microbiological activity in the upper soil layers. The concentration of organic phosphorus was consistently higher in the lysimeter solution than in the suction-cell solution.

If leaching follows preferential paths and the lysimeter solution is derived mainly from the mobile fraction of soil solution as opposed to the suction solution, the observed differences between the methods can be explained. The logarithmically transformed ratios of flux concentrations in lysimeter solution vs. suction solution are compared (Table 3). This reflects the reactivity of chloride, phosphate and the group of compounds determined as organic phosphorus in the soil environment. For all three cases, *t*-tests indicate that the mean values of the logarithmically transformed ratios are significantly different from zero. A negative mean value was attained for chloride, i.e. the chloride concentration was significantly lower in lysimeter solution than in suction-cell solution. In the case of inorganic P and the organic P compounds, the mean logarithmically transformed ratio was positive. Furthermore, the mean value of the ratios of inorganic P was more than

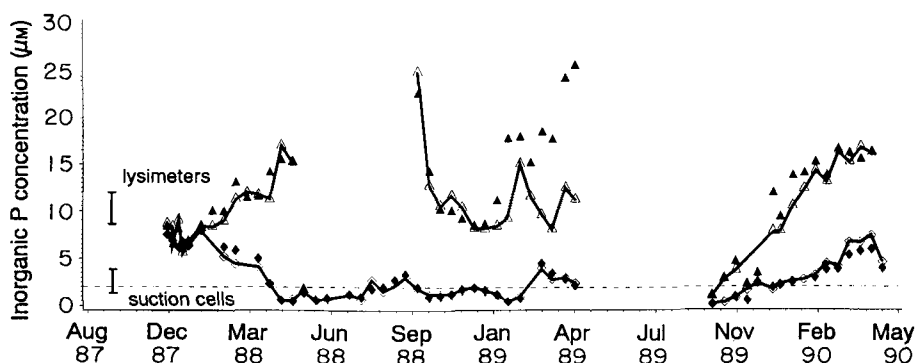


Fig. 3. Inorganic phosphorus in soil solution sampled at 90-cm depth with suction cells and gravitational lysimeters during the period 1987–1990. The broken line indicates the estimated maximum inorganic P concentration for stagnant water at 90 cm. Observed data: open symbols are mean concentrations, solid symbols are flux-averaged concentrations. $\diamond$ $\blacklozenge$, suction cell solution; $\triangle$ $\blacktriangle$, lysimeter solution. An average standard error is shown for both suction cells and lysimeters.

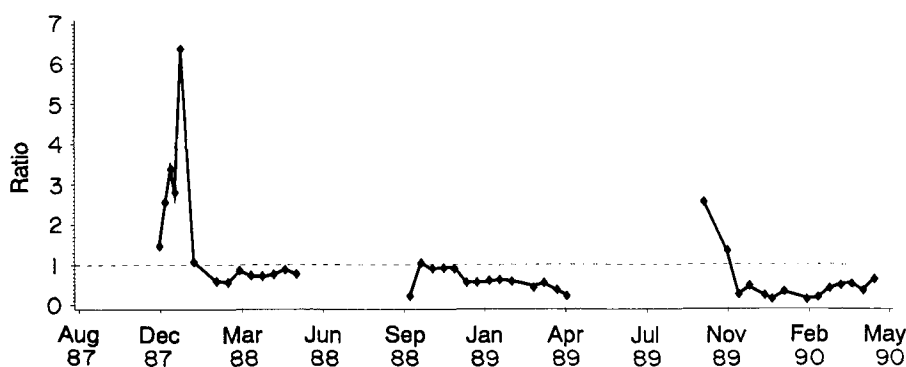


Fig. 4. Ratios of chloride concentrations (lysimeter:suction cell) in soil solution sampled at 90-cm depth with suction cells and gravitational lysimeters during the period 1987–1990.

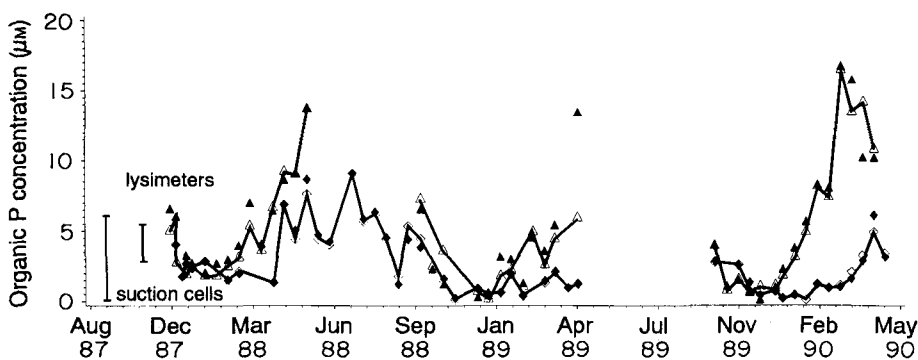


Fig. 5. Organic phosphorus in soil solution sampled at 90-cm depth with suction cells and gravitational lysimeters during the period 1987–1990. Observed data: open symbols are mean concentrations, solid symbols are flux-averaged concentrations. $\diamond$ $\blacklozenge$, suction cell solution; $\triangle$ $\blacktriangle$, lysimeter solution. An average standard error is shown for both suction cells (left bar) and lysimeters (right bar).

Table 3. Descriptive statistics on the solute ratios of lysimeter and suction cell solution

	<i>n</i>	Absolute values			Logarithm mean	<i>T</i> ^b	<i>P</i> > <i>T</i> ^c	<i>L</i> ₉₅ ^d	<i>U</i> ₉₅
		Low	Median	High					
Chloride	42	0.13	0.59	6.36	-0.44	-3.4	0.002	0.624	0.667
Inorganic P	41	0.91	4.98	34.56	1.53	10.1	0.0001	4.34	4.8
Organic P	34	0.13	1.68	16.74	0.67	3.44	0.002	1.82	2.12

^aIn order to compare ratios, logarithmic transformation of ratios (ln) was performed. In all three cases the transformed data conformed closely to a normal distribution; ^b*T*, the Student's *t*-value for testing the hypothesis that the population mean is zero; ^c*P* > *T*, the probability of a greater absolute value for the *t*-value; ^d*L*₉₅ and *U*₉₅ are respectively the lower and upper 95% confidence limits for the mean value of the ratio, based on the transformed values, but transformed back to a linear scale.

twice the ratio for dissolved organic compounds, and this difference is highly significant. This indicates that the forms of organic P dissolved in the plough layer are sorbed by the deeper soil layers to a lesser degree than inorganic P.

It may be argued that the preferential flow paths were merely artefacts created during the process of installation of the lysimeters. Indeed, it was observed that air channels could be formed at the interface between the soil cylinder and perspex tube during transfer of soil from the stainless steel sampling core. This was the result of a small pebble getting stuck at the top of the tube during transfer. For this reason we attempted to follow the movement of soil solution through soil with Rhodamine WT dye. An example of the dispersion of Rhodamine dye is given in Fig. 6. The

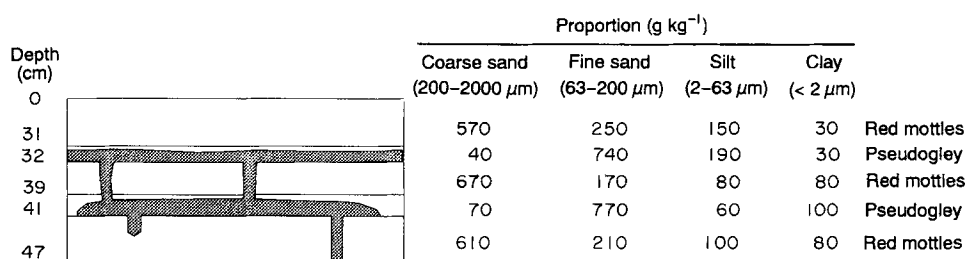


Fig. 6. Schematic representation of the dispersion of Rhodamine dye through alternating layers of varying texture.

observations indicate the existence of preferential flow paths: preferential flow was initiated at the borders of distinct layers, where the upper layer was rich in fine sand having a pseudogley character, and the lower layer had a coarser texture and red mottles. Colour was evenly distributed in the pseudogley layer, but only a small fraction of the lower layer was coloured. In agreement with this, Glass *et al.* (1989) and Baker & Hillel (1990) reached the conclusion that fingering is observed when water passes from a layer with low hydraulic conductivity to layers with higher conductivity. The form of the fingers was shown to be a function of velocity. Glass *et al.* (1989) demonstrated the persistence of fingers formed in layered materials. Thus even after the media were evenly moistened, the main part of the flux was found to take place in the fingers formed at the beginning of the wetting experiment.

Water moving downwards in this characteristic way will reside for a prolonged period in the reduced pseudogley layer, where the sorption of P is slight, and travel quickly through a fraction of the more oxidized layer, where the adsorption capacity for phosphate is more marked. If the flow

paths in the coarser layers are stable over time, as indicated by the work of Glass *et al.* (1989), the adsorption capacity will decrease with time, as the conductive matrix becomes more saturated with regard to phosphate. The observation that flux-averaged concentrations of inorganic P in lysimeter water increased during all the drainage seasons is thus in agreement with stable flow paths. The process of saturation will be faster as the fraction of the matrix involved in the flux decreases.

The difference between the methods most probably occurs because the suction cells tend to extract stagnant soil solution, as the majority of the solution is collected using a considerable pressure gradient. This feature is particularly pronounced in the present case, where vacuum reservoirs with initial low internal pressure were used. The solution isolated in the lysimeters is by nature mobile, as the only hydraulic gradient was that of gravity, enhanced by wicks at the bottom plate.

CONCLUSIONS

In view of the observed differences between the sampling methods with regard to concentration of inorganic P, and the results from the apparent equilibrium determinations, we must conclude that the suction-cell solution was closer to equilibrium with the soil matrix than the lysimeter solution. Application of Rhodamine dye revealed preferential flow paths of leaching water, indicating that the mobile soil solution could be far from equilibrium with the surrounding soil. The difference between methods of extraction was also pronounced for the organic phosphorus compounds and chloride, but to a lesser degree. This is because these compounds have a lower affinity for the soil matrix. Therefore on soils that exhibit preferential flow characteristics, the suction method provides the best representation of the stagnant soil solution concentration, but a poor representation of the flux concentration. The lysimeter method is more appropriate for isolating mobile solution. However, precautions must be taken to avoid the creation of preferential flow paths during installation of the lysimeter.

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